How multimodal gen AI transforms investment decisions – evidence from storytelling through crowdfunding

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0.Abstract

Keywords

1. Introduction

Felista, who is a refugee mother of five school-going children, faces multiple economic challenges in Kenya and is seeking a loan to boost her sales and pay school fees for her children. Felista along with thousands of other small business owners solicit funding through online crowdfunding platforms such as Kiva with the hope of connecting with like-minded investors. In these situations, investment decisions of $25 to $1000 depend on how investors interpret signals from the borrower's pitch, which consists of text and images.

While text and images are virtually bringing together borrowers and investors, crowdfunding platforms are yet to use multimodal generative AI (hereafter referred to as Gen AI) for promoting the cause of small business owners. **This prompts us to raise an important question: how does multimodal Gen AI influence the way in which investors interpret signals from borrowers' pitches and make funding decisions?**

Currently, gen AI is rapidly transforming the way firms interact with business environments, serve their customers and develop persuasive content (Grewal, Satornino, Davenport and Guha, 2024, Schwartz Reismann Institute, 2024 and

Hartmann, Exner and Domdey, 2025.) While several mainstream marketers such as Walmart and Unilever are deploying gen AI to craft customer messages, respond to vendors and create chatbots and new products (Grewal et al, 2024 and Cillo and Rubera, 2025), research linking multimodal Gen AI with investor behavior is limited especially in crowdfunding and subsistence market contexts.

Our study that specifically focuses on how multimodal generative AI impacts crowdfunding decisions is one step towards fulfilling the above research gap. We determine how changes in gen AI modalities affect crowdfunding investment decisions through investment simulations. We develop gen AI content through publicly available profiles on the crowdfunding platform Kiva.

Storytelling, which has become a potent marketing strategy for small businesses with the growth of video-based narratives on digital platforms, is a theoretical anchor for our research. Scholars have previously discussed how linguistic elements of videographic crowdfunding pitches on digital platforms can impact funding success (Peng et al, 2022). In entrepreneurial contexts, storytelling, which often emphasizes the struggles of protagonists, conflicts, temporal and spatial elements and narratives, can positively impact crowdfunding outcomes (De Crescenzo, 2022 and Anglin, Reid and Short, 2023.)

Another key theoretical anchor underpinning our research is signalling theory, which sheds light on how decision-makers use signals to reduce uncertainty while making selections among a choice set under conditions of incomplete information (Bergh et al, 2014). Currently, little is known as to how gen AI-based multimodal storytelling makes investors reinterpret signals from business owners. We build further on the above research gaps on storytelling and signalling to ask another fundamental question: **does gen AI disrupt the relationship between storytelling signals and investment decisions?**

Among the core theoretical contributions of our research is that we find a favorable shift in perceived investability of businesses as a result of using multimodal gen AI pitches. In addition, we find a disruption in the relationship between storytelling signals and investment behavior outcomes with the deployment of gen AI. Interestingly, multimodal gen AI weakens the predictive importance of traditional storytelling cues. Our findings make useful contributions to storytelling and signalling theories, as they give insight into how multimodal gen AI reshapes investors' interpretation of signals from small business borrowers in crowdfunding

environments, which are characterized by incomplete information and uncertainty. The above contributions lend contextual and theoretical novelty to our research which intersects marketing, gen AI and investment behavior in subsistence market contexts.

Our paper is organized as follows. We discuss the contextual, methodological and theoretical novelty of our study in our literature review. This is followed by our methodology section, which gives an in-depth analysis of the concepts used, the hypothesis testing, novelty of our generative AI and LLM tools as well as the robustness of our LLM based analysis.

Key points to add : how gen AI is transforming marketing. What is the impact of gen AI on crowdfunding. Research gap on gen ai modalities and crowdfunding – theoretically enriching our work through rhetorical and sustainability signals. Gen AI and storytelling.

Perhaps we don't dilute all the insights from sustainability and rhetoric – even though these variables do not significantly affect investability perceptions these are interesting contextual variables. For example, we can highlight how emotional appeal/social sustainability became more amplified as opposed to logical reasoning and economic sustainability.

Potential points for managerial implications:

Gen AI modalities and the future of marketing

Gen AI modalities and the future of sustainability marketing

rhetorical and sustainability signals are unreliable indicators of investment decisions

Gen AI, which demonstrates a growing propensity to capture emotional cues, is not only be relevant for developing emotive pitches, but also saving marketing costs, especially for small businesses and social enterprises with a limited marketing budget.

Gen AI and crowfunding
Future research on new modalities

Pointers from reviewers (devil's advocate):
What if a crowdfunding organization can curate videos featuring real folks? Where is the authenticity)?
We can respond to this point saying gen AI videos can be scaled up especially when 1000s of videos featuring
real borrowers cannot be made especially in socio-culturally diverse/sensitive contexts (certain borrowers may be uncomfortable being featured in videos). Also, video quality may vary as would rhetorical/sustainability messaging. Gen AI ensures consistency.

Methodology

2.1. Data Acquisition and Sampling

To empirically establish the foundation of our work, we use a large-scale dataset of microfinance borrower profiles drawn from Kiva.org, a peer-to-peer lending platform operating across more than 80 countries, accessed through the Kiva Developer GraphQL API. The full corpus spans approximately 2.9 million loan records containing textual narratives, photographic imagery, funding timelines, and categorical metadata. To establish this thematic focus, we retain only loans carrying the 'Sustainable Agriculture' tag within Kiva's internal classification system, reducing the corpus to 46,964 records.

Then, we apply stratified random sampling, stratified by borrower country and economic sector. So, the final analytic sample of 1,360 loans, each with complete data across all primary fields, serves as the foundation for every subsequent analytical step.

2.2. Data Processing and Feature Extraction

Having established a representative, thematically focused sample, we construct a unified multimodal representation of each loan, which includes:

Table 1: Multimodal Feature Representation of Loan Data

Variable	Description	Role
Textual Content	Loan narrative (translated)	Main predictor
Visual Content	Listing images	Predictor
Funding Duration	Days to full funding	Outcome
Borrower Country	Borrower location	Covariate
Economic Sector	Standardized sector	Control
Loan Purpose	Standardized purpose	Control

Each record undergoes validation and cleaning to remove entries with missing or inconsistent values; textual data are normalized through encoding standardization and HTML tag removal; and all five attribute categories of Table 1 are aligned into a common schema. With this structured foundation in place, the next step is to design the experimental framework that determines how the data are translated into participant-facing stimuli.

2.3. Experimental Design and Modality Conditions

Building on this dataset, we design a between-subjects experiment in which University marketing students evaluate Kiva-style borrower profiles presented in one of three distinct multimodal configurations. The rubric's design principle is message-content equivalence: loan purpose, requested amount, demographic context. This ensures that any observed variance in participant responses is attributable solely to modality format for causal identification.

The Control condition replicates the current Kiva platform interface, a static borrower photograph paired with the written narrative, establishing the rhetorical baseline against which all treatment effects are assessed. Treatment A adds an AI-generated 12-second video clip alongside the existing text and image, testing the additive effect of synthetic video over the conventional format. Treatment B replaces the static photograph with an AI-generated video while retaining the narrative.

Video stimuli for Treatments A and B are synthesized using Sora 2 (OpenAI, 2025), a diffusion-based text-to-video model that enables precise control over visual scene composition via natural language prompting. For each borrower profile, a standardized prompt is derived directly from the borrower's own narrative; a common template structure that eliminates variation in content quality as a confound. Sora 2 was chosen above alternatives because, at $0.10 per second, it is economically viable to apply profile-specific visual stimuli across 1,360 profiles within our research budget.

2.4. Measurement Instrument

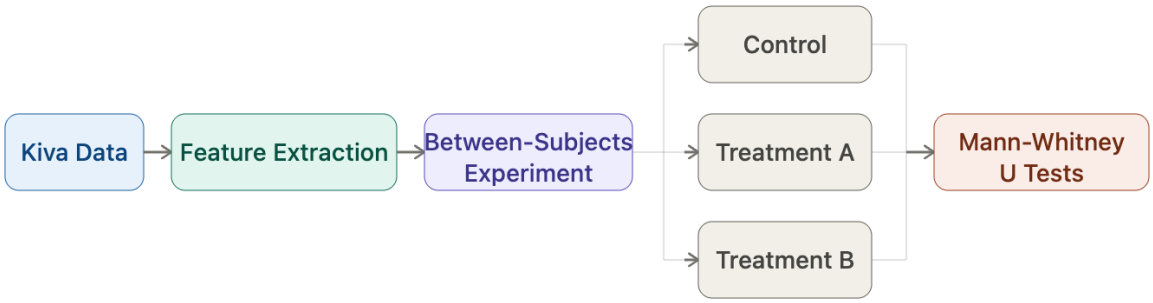


Figure 1: Three-Condition Experimental Framework for Assessing AI-Generated Video in Microfinance

As mentioned in Figure 1, to capture participant responses consistently across all three conditions, we administered a structured 3 × 5 perception grid of 15 ordinal items, each rated on a 4-point Likert scale (1 = Highly Ineffective, 4 = Highly Effective), which is stated below:

Table 2: Multidimensional Perception Grid for Sustainability-based Rhetorical Effectiveness

Dimension	Social	Economic	Environmental
Credibility	Trusted community impact	Proven financial responsibility	Commitment to environmental care
Emotional Appeal	Empathy for vulnerable groups	Hope for economic uplift	Concern for ecological well-being
Logical Reasoning	Equity-driven social improvement	Income growth and stability rationale	Long-term sustainability justification
Timing	Urgent community need	Timely investment opportunity	Urgency of environmental action
Clarity	Clear social benefit articulation	Transparent financial purpose	Clear environmental impact statement

In addition to the 15 perception items of Table 2, participants indicated their investment by selecting 25%, 50%, 75%, or 100% of the borrower's requested amount, which serves as the primary behavioral outcome.

2.5. Inferential Framework and Hypothesis Testing

Building on the 15 perception items and the investment allocation measure described above, we compared all outcomes using one-sided Mann–Whitney U tests instead of parametric tests. This choice was made because the data come from 4-point Likert scales, which are ordinal and do not follow a normal distribution, as is common in crowdfunding evaluation studies. Since Likert data cannot be treated as interval-level measurements, a non-parametric test is more appropriate. We used a one-sided test because our theory predicts a specific direction—that exposure to the treatment will improve borrower perceptions and increase willingness to invest, rather than simply causing any difference.

H_0: Participants in the treatment condition do not rate the borrower more favorably or invest more than those in the control condition.

H_1: Participants in the treatment condition rate the borrower more favorably and invest more than those in the control condition.

Encompassing Control versus Treatment A, Control versus Treatment B, and Treatment A versus Treatment B, Holm–Bonferroni step-down correction is applied across all 16 outcome variables, with the 48 comparisons (16 variables × 3 contrasts) forming a single correction family to preserve the familywise error rate.

2.6. Predicting Funding Outcomes from Perception Scores

While the between-subjects experiment establishes the causal effect of modality on participant perceptions, it cannot, by itself, reveal which of the sustainability-based rhetorics matters most. To answer this question, we administer the same 15-item instrument to 136 participants in the Control condition (Text + Image only). Each participant evaluates 10 distinct borrower profiles from the dataset. Keeping the validation sample to the Control ensures perception scores match the baseline modality of the historical data, avoiding modality-related bias in prediction. Each participant's 15-variable response vector is then paired with the actual funding duration of their assigned Kiva profiles.

These paired records are used to train a Random Forest regression model with funding duration as the outcome variable. Random Forest works well here because it captures nonlinear patterns, handles correlated predictors without extra preprocessing, and highlights the most important features. We train it using 5-fold cross-validation with grid search and evaluate performance on held-out data using RMSE and R^2. The top features are then used for dimension reduction and to guide the design of the enhanced video stimulus described next.

2.7. Video Enhancement via Targeted Prompts

Using the top features identified by the Random Forest model, we design an enhanced prompt leveraging GPT-4o to produce video stimuli. Unlike the neutral prompts used in Treatments A and B, this version adds targeted visual and narrative cues for the most important feature while keeping the same base Sora 2 template.

2.8. Enhancement Effect and Hypothesis Testing

Building on the baseline comparisons in Section 5, this section tests whether prompt-engineered video content leads to stronger borrower evaluations and lending intent than standard AI-generated video. We compare Treatment A and Treatment C, which both use the same modalities (image, text, and video) and differ only in the prompt design used to generate the video. Therefore, any difference between the two can be attributed to the quality of the AI-generated content rather than to differences in media types.

H_0: Participants exposed to the prompt-engineered video (Treatment C) do not rate the borrower more favorably or invest more than those exposed to the standard AI-generated video (Treatment A).

H_1: Participants exposed to the prompt-engineered video (Treatment C) rate the borrower more favorably and invest more than those exposed to the standard AI-generated video (Treatment A).

The 16 Treatment A versus C comparisons form a separate Holm–Bonferroni correction family, reflecting the conceptually distinct inferential question about content quality rather than modality presence, preserving the familywise error rate.

Results

Across all conditions, a clear and consistent pattern emerges exposure to AI-generated video significantly increases participants' willingness to invest. We report this pattern in:

3.1. Modality Effects on Perception and Investment Intent



Figure 2: Changes in Perception Across Sustainability Dimensions Under Treatment Conditions

Figure 2 presents mean score differences on the 3 × 5 perception grid specified in **section 2.4**. Across both Treatment A and Treatment B, Logical Reasoning declines across all three sustainability dimensions here, most sharply in Economic Logical Reasoning (Treatment B: $\Delta = -0.22$) and Environmental Logical Reasoning (Treatment B: $\Delta = -0.18$), while Social Emotional Appeal registers the only consistent positive movement (Treatment A: $\Delta = +0.16$; Treatment B: $\Delta = +0.20$).

Despite these directional patterns, none of the 15 perception items reached statistical significance after Holm–Bonferroni correction in either treatment comparison. The sole significant result is the

primary behavioral outcome Investment Proportion Allocated (derived from **Section 2.4** as a discrete choice among 25%, 50%, 75%, and 100%), which is significantly higher in both treatments: Treatment A ($p < .001$; $\delta = +0.21$, small) and Treatment B ($p < .001$; $\delta = +0.35$, medium).

Table 2: Mann–Whitney U Test Raw p-values for Perception Items and Investment Allocation Across Treatment Conditions

Item	Criterion	Control vs Treatment A	Control vs Treatment B	Treatment A vs Treatment B
Q1	Credibility (Environmental)	0.8702	0.7996	0.3515
Q2	Emotional Appeal (Environmental)	0.1965	0.3658	0.7205
Q3	Logical Reasoning (Environmental)	0.9111	0.9867	0.8044
Q4	Timing (Environmental)	0.1949	0.7313	0.9524
Q5	Clarity (Environmental)	0.9441	0.9395	0.4229
Q6	Credibility (Social)	0.1120	0.1065	0.5273
Q7	Emotional Appeal (Social)	0.0304	0.0067	0.3468
Q8	Logical Reasoning (Social)	0.8783	0.9363	0.6289
Q9	Timing (Social)	0.0699	0.3026	0.8868
Q10	Clarity (Social)	0.7773	0.7478	0.4581
Q11	Credibility (Economic)	0.1709	0.2807	0.6921
Q12	Emotional Appeal (Economic)	0.1044	0.2067	0.7283
Q13	Logical Reasoning (Economic)	0.9827	0.9969	0.6921
Q14	Timing (Economic)	0.0783	0.2172	0.8020
Q15	Clarity (Economic)	0.7839	0.9546	0.7909
Invest	Proportion Allocated (%) (Investment Intent)	0.0001	0.0000	0.0012

Figure 3 illustrates the magnitude of this behavioral shift. The proportion of participants allocating 75% or more rises from 37.6% in the Control to 49.6% in Treatment A and 64.2% in Treatment B, with the 100% allocation share increasing from 10% to 23% and 32%, respectively. Treatment B, which, per **Section 2.3,** replaces the static image entirely with AI-generated video, produces the larger effect, suggesting that substituting rather than supplementing the static photograph yields a stronger behavioral response.

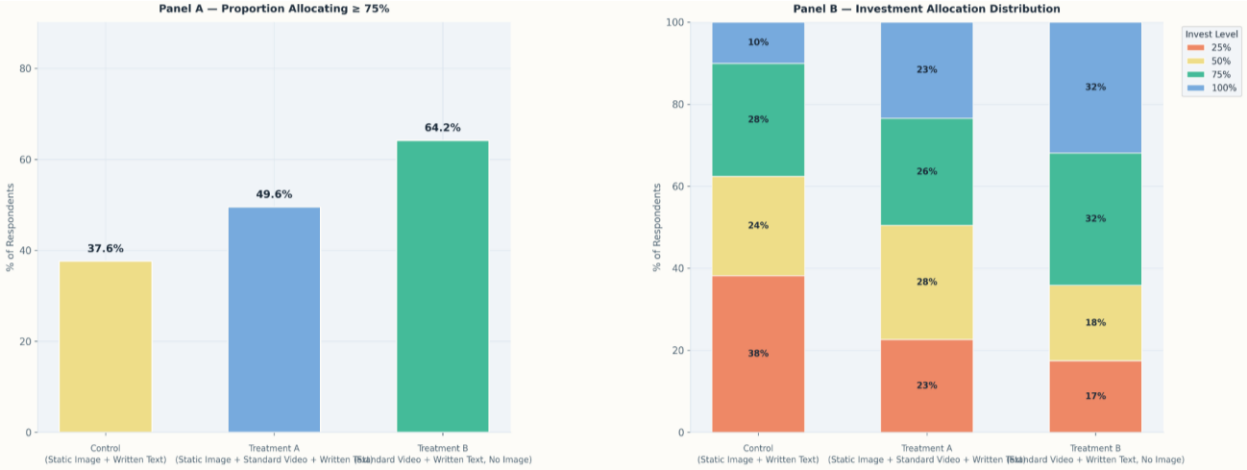


Figure 3: Investment allocation distributions showing increasingly high-allocation responses.

3.2. Predicting Funding Outcomes from Perception Scores: Random Forest Results

For testing the enhancement effect of prompt-engineered video, we report the results of the Random Forest regression described in **Section 2.6**, which pairs perception scores with the actual funding duration of those profiles to identify which rhetorical features most strongly predict real-world lending outcomes. The model is evaluated using 5-fold cross-validation with grid search, and Figure 4 presents both overall model performance and the ranked feature importances.

Across 1,000 cross-validation iterations, the model achieves a mean R^2 of 0.762 (SD = 0.032, median = 0.766, IQR = 0.041) (Figure 4, left panel), with 90% of iterations falling between R^2 = 0.703 and R^2 = 0.806 and an absolute maximum of 0.831. The low coefficient of variation (4.15%) and modest positive skew (0.717) confirm that predictive performance is both strong and stable across resampling folds, indicating that the 15-item perception grid of **Section 2.4 (Table 2)** reliably explains approximately 76% of the variance in actual Kiva funding duration under Control-condition modality.

The Gini-based feature importance ranking (Figure 4, right panel) reveals a clear and stable hierarchy across iterations. **Logical Reasoning (Economic Sustainability)** is the most important predictor (mean = 0.132, SD = 0.009, IQR = 0.012), followed by **Emotional Appeal (Environmental Sustainability)** (mean = 0.115, SD = 0.008, IQR = 0.011) **Credibility (Social Sustainability)** (mean = 0.105, SD = 0.008, IQR = 0.010), Timing (Environmental Sustainability) (mean = 0.101) and Emotional Appeal (Social Sustainability) (mean = 0.075). At the lower end, Timing (Economic Sustainability) contributes least to the model (mean = 0.021, SD = 0.005), with

Timing (Social Sustainability) (mean = 0.033) and Clarity (Economic Sustainability) (mean = 0.036) also ranking among the weakest predictors. Notably, the narrow IQRs across all 15 features, none exceeding 0.012, confirm that the importance rankings are highly stable across resampling folds and are not an artifact of any particular train–test split. These top five features directly informed the GPT-4o-based prompt engineering strategy for Treatment C, as described in **2.6 (Video Enhancement via Targeted Prompts)**.

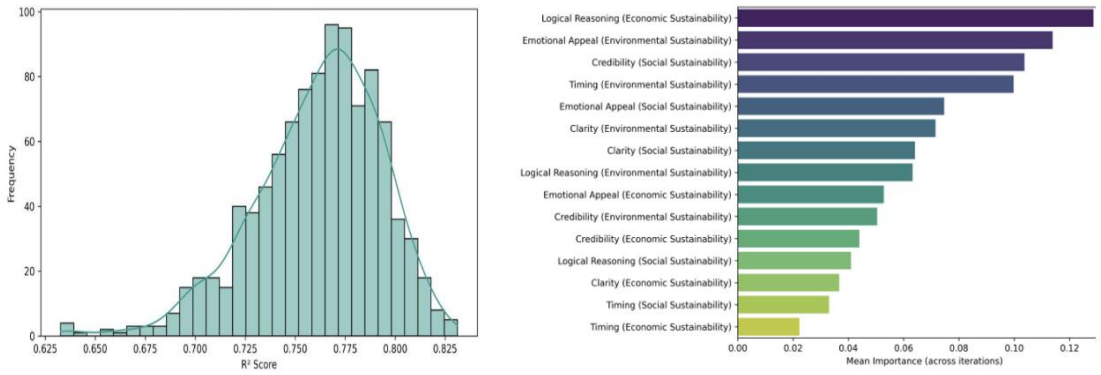


Figure 4:Random Forest Model's overall performance and Gini Feature Importance

3.3. Enhancement Effect: Prompt-Engineered Video (Treatment C vs. Treatment A)

Figure 5 shows the mean score differences between Treatment C and the Control. The pattern replicates that of Section 1: Logical Reasoning declines across all three dimensions (Environmental: $\Delta = -0.29$; Social: $\Delta = -0.13$; Economic: $\Delta = -0.32$) and Social Emotional Appeal again shows the strongest positive movement ($\Delta = +0.25$). The prompt-engineered video generated via GPT-4o–refined Sora 2 prompts targeting the top Random Forest predictors identified in **Section 2.6** does not resolve the perception-level null pattern observed for Treatments A and B.

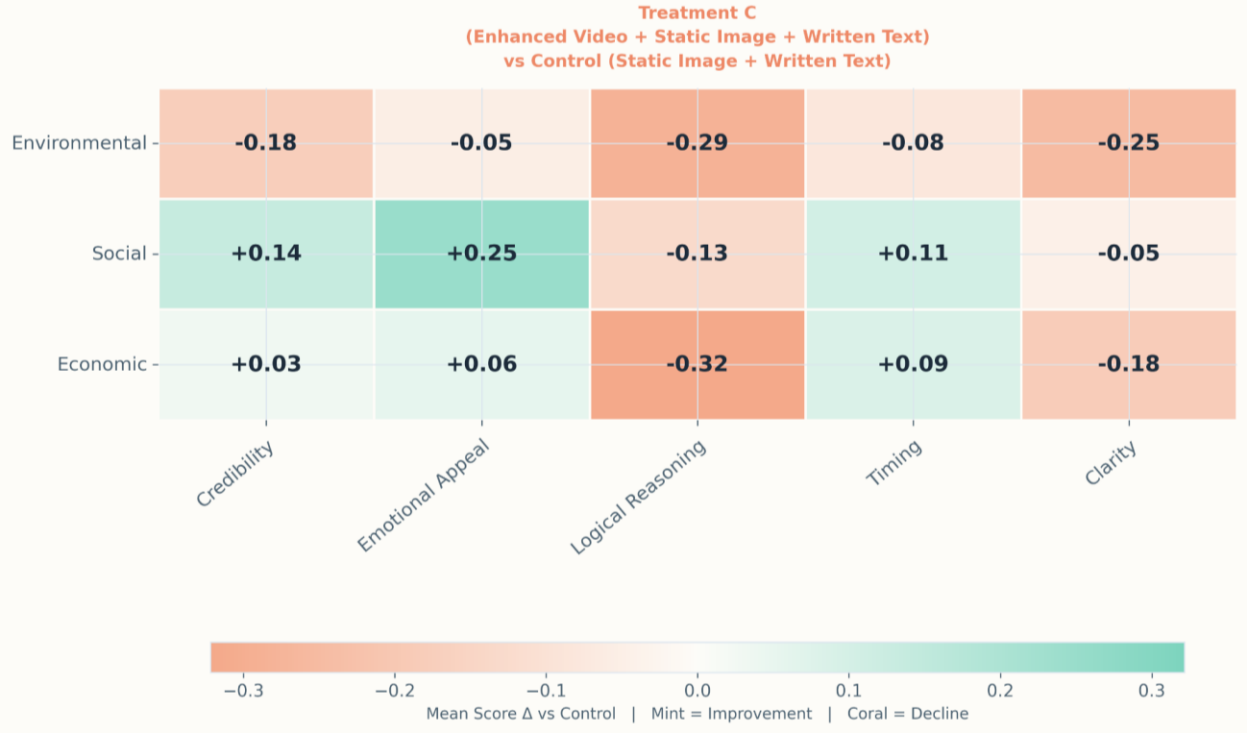


FIGURE 5: Mean Score Difference: Control &Treatment C (Enhanced Video + Image + Written Text)

Behaviorally, however, Treatment C produces the strongest investment response of all conditions: 67.8% of participants allocate 75% or more (vs. 37.6% in Control), and the 100% allocation share rises from 10% to 48%, the highest full-allocation rate across the study. Figure 6 presents the full distribution.

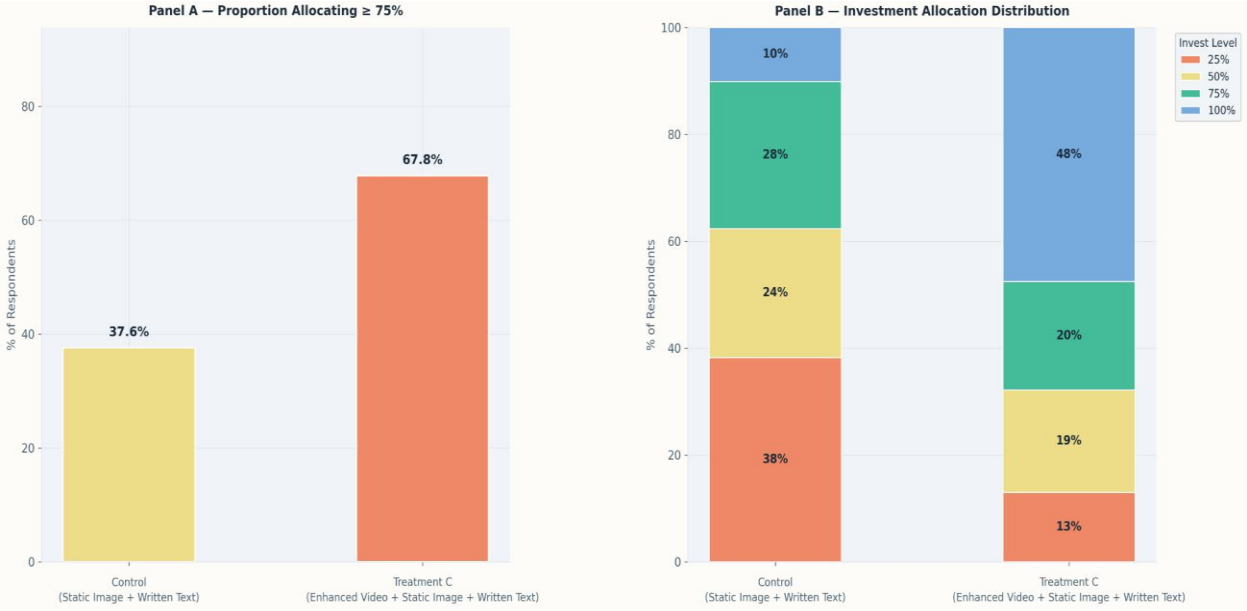


FIGURE 6: Investment Allocation: Control vs Treatment C

The critical comparison, specified as a separate correction family in **Section 2.7**, is Treatment C vs. Treatment A — conditions identical in modality and differing only in video prompt quality. As shown in Table 3, none of the 15 perception items differ significantly after Holm–Bonferroni correction (all $|\delta| \leq 0.10$; negligible). Investment Proportion Allocated, however, is significantly higher in Treatment C ($p < .001$; $\delta = +0.27$, small), directly supporting H_1 of **2.7** and confirming that prompt engineering adds behavioral value beyond standard AI-generated video, independently of modality format.

Table 3: Mann–Whitney U Test Raw p-values for Perception Items and Investment Allocation for Treatment C

Item	Criterion	Treatment A vs Treatment C
Q1	Credibility (Environmental)	0.8661
Q2	Emotional Appeal (Environmental)	0.9530
Q3	Logical Reasoning (Environmental)	0.9858
Q4	Timing (Environmental)	0.9795
Q5	Clarity (Environmental)	0.8398
Q6	Credibility (Social)	0.3646
Q7	Emotional Appeal (Social)	0.1484
Q8	Logical Reasoning (Social)	0.6230
Q9	Timing (Social)	0.6773
Q10	Clarity (Social)	0.3096
Q11	Credibility (Economic)	0.5946
Q12	Emotional Appeal (Economic)	0.6383
Q13	Logical Reasoning (Economic)	0.9770
Q14	Timing (Economic)	0.6816
Q15	Clarity (Economic)	0.9144
Invest	Proportion Allocated (%) (Investment Intent)	0.0000